

Name: _____

Part One- Multiple Choice**40 Points**

1. If $f(x) = x^4 + x^3 + 1$ then $\lim_{w \rightarrow 1} \frac{f'(w) - f'(1)}{w - 1} =$
 A. 36 B. 9 C. -3 D. 24
2. $\lim_{x \rightarrow 1} \frac{\sin(x^2 - 1)}{x - 1} =$
 A. 1 B. 2 C. 3 D. 4
3. The slope of the tangent to the curve of $f(x) = \sin x + 2x$ when $x = \pi$ is equal to
 A. 3 B. 2 C. 1 D. 0
4. If $f(3) = -2$, $f'(3) = 4$ and $g(x) = \frac{2x+1}{f(x)}$ then $g'(3) =$
 A. 6 B. 8 C. -6 D. -8
5. If $f(x) = 2x - 1$ then $f^{-1}(x) =$
 A. $\frac{x-1}{2}$ B. $\frac{x+1}{2}$ C. $x+2$ D. $x=2$

6. Given that $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 2$ then $\frac{dy}{dx}$ at the point (1, 1) equals
 A. 0 B. 2 C. 1 D. -1

7. At $x = 5$ the function, $f(x) = \sqrt[3]{x-5}$ has
 A. a vertical tangent C. a relative maximum
 B. a cusp D. a relative minimum

8. The function $f(x) = x(x-4)^3$ is increasing on
 A. $(-\infty, \infty)$ B. $(0, \infty)$ C. $[1, \infty)$ D. $(-\infty, 1]$
9. The function, $f(x) = x|x-2|$ is concave up on
 A. $(-\infty, \infty)$ B. $(0, \infty)$ C. $(2, \infty)$ D. $(-\infty, 2)$

A. 0

B. 2

C. 1

D. -1

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A. $(-\infty, \infty)$ B. $(0, \infty)$ C. $(2, \infty)$ D. $(-\infty, 2)$

10. $\int (1 + \sin^4 \theta \csc \theta) d\theta =$

A. $\theta + \sin \theta + c$ C. $\theta + \cos \theta + c$ B. $\theta - \sin \theta + c$ D. $\theta - \cos \theta + c$

11. Given that $\int_0^x f(x) \sin^4 x \, dx = 5$ and $\int_0^x f(x) \cos^3 x \, dx = 7$ then $\int_0^b f(x) \, dx =$

A. 5

B. -2

C. 7

D. 12

12. $\frac{d^2}{dx^2} (x^3 + x^2 + 2) =$

A. $3x^3 + 2x$ B. $6x + 2$ C. $6x + 1$

D. 6

13. $\int_0^x \frac{x}{\sqrt{x^3 + 1}} =$

A. $\sqrt{17} - 1$ B. $\sqrt{17}$ C. $\sqrt{17} - \sqrt{2}$ D. $\sqrt{17} - 2$

14. The area of the region in the first quadrant, which is bounded by the coordinate axes and the curve of the function $f(x) = \cos x$ is equal to

A. 2 square units

B. 1.5 square units

C. 2.5 square units

D. 1 square unit

Hint: Rints $x = 0$ to $a = 1$

15. The arc length of the curve of the function $f(x) = 2\sqrt{x^3 + 1}$ from $x = 0$ to $x = 1$
- A. $\frac{2}{27}(10\sqrt{10} + 1)$ B. $\frac{2}{27}(19\sqrt{19} + 1)$ C. $\frac{2}{27}(10\sqrt{10} - 1)$ D. $\frac{2}{27}(19\sqrt{19} - 1)$
16. The relative minimum of the function $f(x) = \frac{1}{3}x^3 - 3x^3 + 5x + 10$ occurs, when $x =$
- A. 3 B. 1 C. 2 D. 5

1) Compute: $\int_0^1 \frac{\sin^4 x}{\cos x} dx$

(5 points)

- 2) Given that $f(x) = \begin{cases} 2x & , 0 \leq x \leq 1 \\ 2 & , 1 < x \leq 4 \end{cases}$. Find $F(x) = \int_0^x f(t) dt$ as a piecewise function. (5 points)

①

$$* \lim_{w \rightarrow 1} \frac{\hat{f}(w) - \hat{f}(1)}{w-1} = \hat{f}'(1)$$

$$\hat{f}(x) = 30x^4 + 6x$$

$$= 30 + 6 = 36 \quad \text{A}$$

②

$$* \lim_{x \rightarrow 1} \frac{\sin(x^2-1)}{x-1} \times \frac{(x+1)}{x+1} \rightarrow \lim_{x \rightarrow 1} \frac{\sin(x^2-1)}{x^2-1} \times (x+1)$$

$$= 1 \times 2 = 2 \quad \text{B}$$

③

$$* \hat{f}(x) = \cos(x) + 2$$

$$f(x) = \sin x + 2x$$

$$\hat{f}(\pi) = \cos(\pi) + 2 = 1 \quad \text{C}$$

④

$$* f(3) = -2 \quad \hat{f}(3) = 4$$

$$g(x) = \frac{2x+1}{f(x)}$$

$$g(x) = \frac{2f(x) - \hat{f}(x)(2x+1)}{f^2(x)}$$

$$g(3) = \frac{2(-2) - 4 \times 7}{4} = \frac{-4 - 28}{4} = -8 \quad \text{D}$$

⑤ $f(x) = 2x-1$

$\hat{f}^{-1}(x) = \frac{x+1}{2}$ (B)

⑥ $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 2$ at $(1,1)$

$$\frac{2}{3x^{\frac{1}{3}}} + \frac{2 \frac{dy}{dx}}{3y^{\frac{1}{3}}} = 0 \rightarrow \frac{2}{3} + \frac{2 \frac{dy}{dx}}{\frac{2}{3}} = 0$$

$\rightarrow \frac{dy}{dx} = -1$ (D)

⑦ (A) a vertical tangent

⑧ $f(x) = x(x-4)^3$

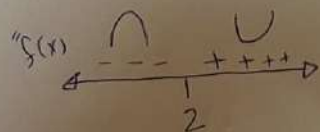


$[1, \infty)$ (C)

$$\begin{aligned} \hat{f}(x) &= 3x(x-4)^2 + (x-4)^3 \\ &= (x-4)^2(3x+x-4) \\ &= (x-4)^2(4x-4) \\ &= (x-4)^2(4x-4) \\ &= (x-4)^2(4x-4) \end{aligned}$$

$x=4 \quad x=1$

⑨ $f(x) = x|x-2|$



$(2, \infty)$ (C)

$$f(x) = \begin{cases} x(x-2), & x \geq 2 \\ x(2-x), & x < 2 \end{cases}$$

$$\hat{f}(x) = \begin{cases} 2, & x \geq 2 \\ -2, & x < 2 \end{cases}$$

$$\textcircled{10} \int (1 + \sin^2 \theta \csc \theta) d\theta = \theta - \cos \theta + C$$

(D)

$$\textcircled{11} \int_a^b f(x) \sin^2 x \, dx = 5 \quad \int_a^b f(x) \cos^2 x \, dx = 7$$

$$\int_a^b f(x) \sin^2 x \, dx + \int_a^b f(x) \cos^2 x \, dx = 12$$

$$\int_a^b f(x) (\sin^2 x + \cos^2 x) \, dx = 12$$

$$\int_a^b f(x) \, dx = 12 \quad \#$$

$$\textcircled{12} \frac{d^2}{dx^2} [x^3 + x^2 + 2] = 3 \cdot 2x^1 + 2 \cdot 1x^0 \\ = 6x + 2$$

(B) $6x + 2$

$$\textcircled{13} \int_0^4 \frac{x}{\sqrt{x^2+1}} = \int_1^{17} \frac{x}{\sqrt{u}} \times \frac{du}{2x} \quad \begin{array}{l} u = x^2 + 1 \\ du = 2x \, dx \end{array}$$

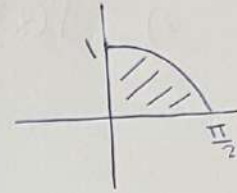
$$\frac{1}{2} \int_1^{17} \frac{1}{\sqrt{u}} \, du = \frac{1}{2} \left. 2u^{\frac{1}{2}} \right|_1^{17} = \sqrt{17} - 1$$

(A)

(14) $f(x) = \cos x$

$$\int_0^{\frac{\pi}{2}} \cos x = \sin x \Big|_0^{\frac{\pi}{2}}$$

$$= 1$$



(D)

(15) $f(x) = 2\sqrt{x^3} + 1$

$x=0$ to $x=1$

$$L = \int_0^1 \sqrt{1 + (f'(x))^2} dx$$

$f'(x) = 3\sqrt{x}$

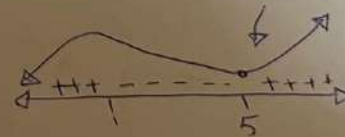
$$L = \int_0^1 \sqrt{1 + (3\sqrt{x})^2} dx = \int_0^1 \sqrt{1 + 9x} dx$$

$$L = \frac{2}{27} (9x+1)^{\frac{3}{2}} \Big|_0^1 = \frac{2}{27} (10\sqrt{10} - 1) \neq$$

(C)

(16) $f(x) = \frac{1}{3}x^3 - 3x^2 + 5x + 10$

at $x=5$



(D)

$$f'(x) = x^2 - 6x + 5$$

$$= (x-1)(x-5)$$

Q₂)

$$2) f(x) = \begin{cases} 2x & , 0 \leq x \leq 1 \\ 2 & , 1 < x \leq 4 \end{cases}$$

$$F(x) = \begin{cases} \int_0^x 2x \, dx, & 0 \leq x \leq 1 \\ \int_0^1 2x \, dx + \int_1^x 2 \, dx, & 1 < x \leq 4 \end{cases}$$

$$\begin{aligned} * x^2 \Big|_0^x &= x^2 \\ * x^2 \Big|_0^1 + 2x \Big|_1^x &= 1 + 2x - 2 \\ &= 2x - 1 \end{aligned}$$

$$F(x) = \begin{cases} x^2, & 0 \leq x \leq 1 \\ 2x - 1, & 1 < x \leq 4 \end{cases}$$

$$= \begin{cases} x^2, & 0 \leq x \leq 1 \\ 2x - 1, & 1 < x \leq 4 \end{cases}$$